

Illuminati - 2019

Solutions to Advanced Physics Assignment -2B Liquids, POM, Thermodynamics, SHM & Waves

SINGLE OPTION CORRECT TYPE

- 1.(C) Vertical component of force of surface tension on liquid $= (2\pi r \cos \theta) \sigma$

Work done by this force during capillary rise $= W = (2\pi \sigma r \cos \theta) \times h$

$$\text{Where } h = \frac{2\sigma \cos \theta}{r\rho g} \Rightarrow W = (2\pi \sigma r \cos \theta) \times \frac{2\sigma \cos \theta}{r\rho g} = \frac{4\pi \sigma^2 \cos^2 \theta}{\rho g}$$

$$\text{Potential energy increase} = P = mg \left(\frac{h}{2} \right) = (\pi r^2 h)(\rho) \left(\frac{gh}{2} \right) = \frac{\pi \rho r^2 gh^2}{2}$$

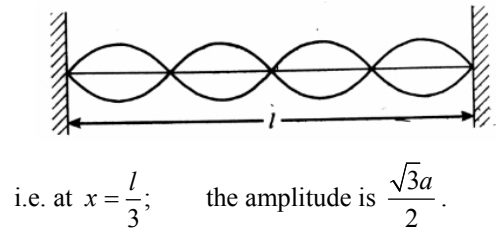
$$\Rightarrow P = \frac{2\pi \sigma^2 \cos^2 \theta}{\rho g} \quad \text{Heat liberated} = W - P = \frac{2\pi \sigma^2 \cos^2 \theta}{\rho g}$$

- 2.(C) For a string vibrating in its n th overtone $[(n+1)^{\text{th}}$ harmonic]

$$y = 2A \sin \left(\frac{(n+1)\pi x}{L} \right) \cos \omega t$$

$$\text{For } x = \frac{l}{3}, 2A = a \text{ and } n = 3; y = \left[a \sin \left(\frac{4\pi}{l} \cdot \frac{l}{3} \right) \right] \cos \omega t$$

$$= a \cdot \sin \frac{4\pi}{3} \cos \omega t = -a \cdot \left(\frac{\sqrt{3}}{2} \right) \cos \omega t$$



- 3.(A) $\Delta V = V \theta \gamma, \frac{\Delta V}{V} = \gamma \theta$

$$\text{Stress} = \left(\frac{\Delta V}{V} \right) B$$

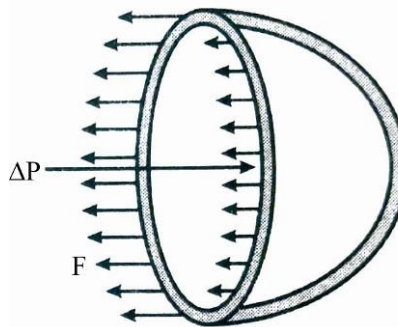
$$\Delta P = \gamma \theta B$$

$$\frac{F}{\pi R^2} = \gamma \theta B$$

$$F = \pi \gamma \theta B R^2$$

$$\text{Tensile stress} = \frac{F}{2\pi R t} = \frac{\pi \gamma \theta B R^2}{2\pi R t}$$

$$\text{Tensile stress} = \left(\frac{\gamma \theta B R}{2t} \right)$$



- 4.(A) Average kinetic energy with respect to space $= \frac{1}{3} k A^2$

$$\text{Average potential energy with respect to space} = \frac{1}{6} k A^2$$

$$\text{Average kinetic energy with respect to time} = \frac{1}{4} k A^2$$

$$\text{Average potential energy with respect to time} = \frac{1}{4} k A^2$$

5.(C) $\lambda = \text{mean free path} = \frac{1}{\sqrt{2}\pi n d^2}$

Where, $n = \frac{N}{V} = \frac{p}{k_B T}$

$$\therefore \lambda = \frac{k_B \cdot T}{\sqrt{2} \cdot \pi \cdot p \cdot d^2} \quad \text{So,} \quad \frac{\lambda_2}{\lambda_1} = \frac{k_B \cdot T}{\sqrt{2} \cdot \pi \cdot p \cdot d^2} \cdot \frac{\sqrt{2} \pi p_1 d^2}{k_B T_1}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{p_1}{p_2} \quad \Rightarrow \quad \lambda_2 = \lambda_1 \cdot \frac{p_1}{p_2} = \frac{9.5 \times 10^{-8} \times p_1}{\left(\frac{p_1}{100}\right)} = 9.5 \times 10^{-6} \text{ m}$$

6.(B) Relative to liquid, the velocity of sphere is $2v_0$ (upwards).

$\therefore$ Viscous force on sphere = $6\pi\eta r(2v_0)$ (downwards) = $12\pi\eta r v_0$ (downwards)

7.(A) Rate of cooling of a body is the ratio of radiation power emitted by body and its heat capacity. As dimensions, surface material and temperature are equal the radiation power of both spheres will be same but heat capacity of hollow sphere is less so it will cool at a faster rate.

8.(C) $dp = -\rho g dh = -\frac{pm}{kT} \cdot g dh$

$$\Rightarrow \int_{p_0}^p \frac{dp}{p} = -\int_0^h \frac{mg}{kT} \cdot dh \quad \Rightarrow \quad p = p_0 e^{-\left(\frac{mg}{kT}\right)h}$$

$$\begin{cases} m_\infty &= \text{mass till infinite length of column of area } A \\ m_h &= \text{mass till height } h \text{ of column of area } A \end{cases}$$

$$p_0 = \frac{m_\infty g}{A}$$

$$p_h = \frac{(m_\infty - m_h)g}{A} = \frac{m_\infty g}{A} e^{-\frac{mgh}{kT}}$$

$$1 - \frac{m_h}{m_\infty} = e^{-\frac{mgh}{kT}}$$

$$\frac{m_h}{m_\infty} = \frac{n_h}{n_\infty} = 1 - e^{-\frac{mgh}{kT}}$$

9.(C) The time period of free oscillation of pendulum $T = 2\pi\sqrt{\frac{l}{g}} = 2s$

Time taken by bob to go from extreme position A to mean position B is $\frac{T}{4}$

Time taken by bob to move from mean position B to position C (where its angular displacement α is half the angular amplitude β) is found from equation

$$\alpha = \beta \sin\left(\frac{2\pi}{T}t\right)$$

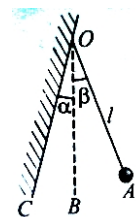
Solving we get $t = \frac{T}{12}$

$$\Rightarrow \text{Total time period of oscillation is} = 2\left[\frac{T}{4} + \frac{T}{12}\right] = \frac{2}{3}T = \frac{4}{3}s$$

10.(D) Path difference $(\Delta x) = 50 \text{ cm} = \frac{1}{2} \text{ m}$

$$\Rightarrow \text{Phase difference due to path difference } \Delta\phi = \frac{2\pi}{\lambda} \times \Delta x \Rightarrow \phi = \frac{2\pi}{1} \times \frac{1}{2} = \pi$$

$$\text{Total phase difference} = \pi + \frac{\pi}{3} = \frac{4\pi}{3} \quad \Rightarrow \quad A = \sqrt{a^2 + a^2 + 2a^2 \cos(4\pi/3)} = a$$



- 11.(D)** Pressure at S is less than atm pressure $P_S = P_0 - \rho g(H - h)$
So, mercury will not come out.
- 12.(C)** As the volume of both are same, according to the thermal expansion concept, the change in volume is proportional to the total volume enclosed by the material hence it will be same for both spheres.
- 13.(C)** $\Delta Q = \Delta U + \Delta W$

$$\Rightarrow \Delta U = \Delta Q - \Delta W = 540 - \frac{p\Delta V}{J}$$

$$\Rightarrow \Delta U = 540 - \frac{1.013 \times 10^5 (1671 - 1) \times 10^{-6}}{4.2} = 540 - 39.7 \approx 500 \text{ cal}$$
- 14.(B)** On rising up as g decreases the time period of the pendulum increases and clock will slow down and start losing time, time lost per unit time can be given as $dT/T = (1/2)dg/g$ here $g = GM/R^2$ and verify that option **(B)** is correct.
- 15.(D)** At $x = 0$ the phase difference should be π for destructive interference
Alternate solution
We use $y_2 = a \cos(\omega t + kx + \phi_0)$
Resultant after superposition is

$$y = y_1 + y_2 = a \cos(\omega t - kx + \frac{\pi}{3}) + a \cos(\omega t + kx + \phi_0) = 2a \cos \left[\omega t + \frac{\pi + \phi_0}{2} \right] \times \cos \left[kx + \frac{\phi_0 - \frac{\pi}{3}}{2} \right]$$

$$\Rightarrow y = 0 \text{ at } x = 0 \text{ for any } t \Rightarrow kx + \frac{\phi_0 - \frac{\pi}{3}}{2} = \frac{\pi}{2} \text{ at } x = 0$$

$$\Rightarrow \phi_0 = \frac{4\pi}{3}. \text{ Hence } y_2 = a \cos(\omega t + kx + \frac{4\pi}{3})$$
- 16.(C)** The maximum static frictional force is $f = \mu mg \cos \theta = 2 \tan \theta mg \cos \theta = 2mg \sin \theta$
Applying Newton's second law to block at lower extreme position, we have $f - mg \sin \theta = m\omega^2 A$

$$\Rightarrow \omega^2 A = g \sin \theta \quad \text{As } \omega = \sqrt{\frac{k}{3m}} \Rightarrow A = \frac{3mg \sin \theta}{k}$$
- 17.(B)** To produce 5 beats/sec. Frequency of one wire should be increase up to 505 Hz. i.e. increment of 1% in basic frequency.

$$n \propto \sqrt{T} \text{ or } T \propto n^2 \Rightarrow \frac{\Delta T}{T} = 2 \frac{\Delta n}{n}$$

$$\Rightarrow \text{percentage change in Tension} = 2(1\%) = 2\%$$
- 18.(C)** As, $\frac{C-0}{100-0} = \frac{N-100}{400-100} \Rightarrow \frac{x}{100} = \frac{x-100}{300}$

$$\Rightarrow x = \frac{x-100}{3} \Rightarrow x = -50^\circ\text{C}$$
- 19.(C)** At 4°C density of water is maximum so when heat is conducted and temperature of water at different regions become 4°C , it settles down at bottom and when temperature further falls below 4°C its density decreases and it will float above so 0°C will attain first at the region in between and at top.
- 20.(A)** For chamber : $\frac{dQ}{dt} = k(\theta_1 - \theta_0) = k(\theta_2 - \theta_0) \Rightarrow \theta_1 = \theta_2$
For heater : $\frac{dQ}{dt} = e_1 A \sigma (T_1^4 - \theta_1^4) = e_2 A \sigma (T_2^4 - \theta_2^4) \Rightarrow \theta_1 = \theta_2$

$$e_1 > e_2 \Rightarrow T_1 < T_2$$

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

21.(AD)

$$v_1 = \sqrt{2gh} \quad v_2 = \sqrt{2g(h+H)}$$

$$x = v_1 t_1 = v_2 t_2 \quad \text{and} \quad y = h + \frac{1}{2} g t_1^2$$

$$\text{and} \quad y = h + H + \frac{1}{2} g t_2^2$$

The point of intersection will remain at the same place if the level of water in the vessel does not change. That means $Q = A_1 v_1 + A_2 v_2$

$$\text{or} \quad Q = a(v_1 + v_2), \quad v_1 + v_2 = \frac{Q}{a}$$

$$\Rightarrow \sqrt{2gh} + \sqrt{2g(h+H)} = \frac{Q}{a}, \quad \sqrt{2g(h+H)} = \frac{Q}{a} - \sqrt{2gh}$$

On squaring both sides, we get

$$2gh + 2gH = \frac{Q^2}{a^2} + 2gh - 2\frac{Q}{a}\sqrt{2gh}$$

$$2\frac{Q}{a}\sqrt{2gh} = \frac{Q^2}{a^2} - 2gH, \quad 2\left(\frac{140}{0.2}\right)\sqrt{2 \times 980 \times h} = \frac{140^2}{(0.2)^2} - 2 \times 980 \times 50$$

$$\Rightarrow h = 40 \text{ cm} \quad \therefore v_1 = \sqrt{2 \times 980 \times h} = 280 \text{ cms}^{-1}$$

$$\text{and} \quad v_2 = \sqrt{2 \times 980(40+50)} = 420 \text{ cms}^{-1}$$

$$t_1 = \frac{v_2}{v_1} t_2 = \frac{420}{280} t_2 = \frac{3}{2} t_2$$

$$\therefore y = 40 + \frac{1}{2} \times 980 \times \frac{9}{4} t_2^2 = 40 + 50 + \frac{1}{2} \times 980 t_2^2$$

$$\frac{1}{2} \times 980 \left(\frac{5}{4} t_2^2 \right) = 50 \quad \Rightarrow \quad t_2 = \left(\frac{4}{49} \right) = \frac{2}{7} \text{ s}$$

$$\therefore t_1 = \frac{3}{7} \text{ s} \quad \therefore x = v_1 t_1 = (280) \left(\frac{3}{7} \right) = 120 \text{ cm}$$

$$\text{and} \quad y = h + \frac{1}{2} g t_1^2 = 40 + \frac{1}{2} (980) \left(\frac{9}{49} \right) = 130 \text{ cm}$$

22.(ABCD)

$$f_0 = \frac{1}{2L} \sqrt{\frac{F}{\mu}} = \frac{1}{2L} \sqrt{\frac{F}{\rho A}}$$

Where, F = tension

μ = mass per unit length

$$\text{Stress} = \frac{F}{A} = Y (\text{Strain}) = Y \left(\frac{\Delta L}{L} \right) = Y \left(\frac{L \alpha \theta}{L} \right) = Y \alpha \theta$$

$$\therefore f_0 = \frac{1}{2L} \sqrt{\frac{Y \alpha \theta}{\rho}} \quad \therefore f_0 \propto \frac{1}{L} \propto \sqrt{Y} \propto \sqrt{\theta} \propto \sqrt{\frac{\alpha}{\rho}}$$

23.(BCD)

Due to the pseudo force on block (considered external) its mean position will shift to a distance mg/K above natural length of spring as net force now is mg in upward direction. So total distance of block from new mean position is $2mg/K$ which will be the amplitude of oscillations hence option (C) is correct. During oscillations spring will pass through the natural length hence option (D) is correct. As block is oscillating under spring force and other constant forces which do not affect the SHM frequency hence option (B) is correct.

24.(ABCD)

By Dulong-petit law, molar specific heat of crystalline solids is $3R$

25.(BD) $r' = r(1 + \alpha\Delta T)$

$d' = d(1 + \alpha\Delta T)$

The total length will also increase $\ell' = \ell(1 + \alpha\Delta T)$

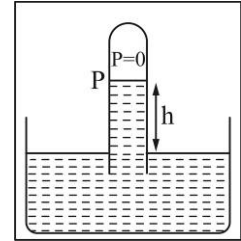
As length and radius increases by same factor, their ratio (angle) remains same.

26.(BCD) When stationary $\rho gh = P$

When accelerating upwards $\rho(g + a)h_1 = P \Rightarrow h_1 < h$

When accelerating downwards $\rho(g - a)h_2 = P \Rightarrow h_2 > h$

When moving with constant velocity $\rho gh_3 = P \Rightarrow h_3 = h$



27.(BD) KE for an ideal gas $= \frac{f}{2} nRT = \frac{5}{2} nRT$

Also, KE of gas w.r.t. ground $= KE$ of COM w.r.t. ground $+ KE$ of gas w.r.t. COM of system

$= \frac{1}{2} mv^2 + \frac{5}{2} nRT$

28.(AC) When the two waves superpose then the maximum amplitude will be $4 + 3 = 7$ and minimum will be $4 - 3 = 1$ hence the ratio of maximum to minimum intensity will be 49 and as the frequencies of the two waves are 200 Hz and 202 Hz the beat frequency will be 2Hz.

29.(AC) Volume decreases during melting of ice.

$\therefore$ Positive work is done on the ice-water system by the atmosphere

Heat is absorbed by ice-water system. So, internal energy increases.

30.(ABC) $\omega' = \sqrt{\frac{k}{m_1}}; v_1 = \sqrt{\frac{k}{m_1 + m_2}} x_0$

$\omega' A' = \sqrt{\frac{k}{m_1 + m_2}} x_0; A' = \sqrt{\frac{m_1}{m_1 + m_2}} x_0$

PARAGRAPH TYPE

31.(B) Since the vessel network is symmetrical, flow in a vessel of level $i + 1$ is half the flow in a vessel of level i .

In this way, we can sum the pressure differences in all levels : $\Delta P = \sum_{i=0}^{N-1} Q_i R_i = Q_0 \sum_{i=0}^{N-1} \frac{R_i}{2^i}$

Introducing the radii dependences yields $\Delta P = Q_0 \sum_{i=0}^{N-1} \frac{8\ell_i \eta}{2^i \pi r_i^4} = Q_0 \frac{8\ell_0 \eta}{\pi r_0^4} \sum_{i=0}^{N-1} \frac{2^{4i/3}}{2^i} = Q_0 N \frac{8\ell_0 \eta}{\pi r_0^4}$

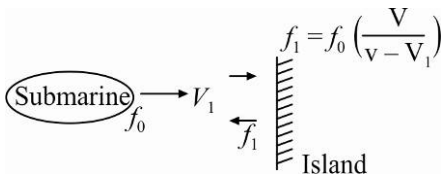
Therefore $Q_0 = \Delta P \frac{\pi r_0^4}{8N\ell_0 \eta}$

Hence, the flow rate for a vessel network in level i is $Q_i = \Delta P \frac{\pi r_0^4}{2^{i+3} N \ell_0 \eta}$

32.(C) Replace values in the formula and change units appropriately

$Q_0 = \frac{\Delta P \pi r_0^4}{8N\ell_0 \eta} = \frac{(55-30) \times 1.013 \times 10^5 \times 3.1415 \times (6.0 \times 10^{-5})^4}{760 \times 48 \times 2.0 \times 10^{-3} \times 3.5 \times 10^{-3}} = 4 \times 10^{-10} \text{ m}^3/\text{s}$

To obtain the final value in the requested units : $Q_0 \approx 1.5 \text{ ml/h}$

33.(B) 
$$f'' = f_0 \left(\frac{V + V_1}{V - V_1} \right), V = 1050 \quad \left[\frac{f'' - f_0}{f_0} = 0.1 \right]$$

$$\frac{f'' - f_0}{f_0} = \frac{2V_1}{V - V_1} = 0.1$$

$V_1 = 50 \text{ m/sec.}$

34.(A) 
$$f'' = f' \left(\frac{V + 50}{V - V_2} \right) \quad f' = f_0 \left(\frac{V + V_2}{V - 50} \right)$$

$$f'' = f_0 \left(\frac{(V + V_2)(V + 50)}{(V - V_2)(V - 50)} \right) = 1.21 f_0 \text{ [21\% greater than sent waves]}$$

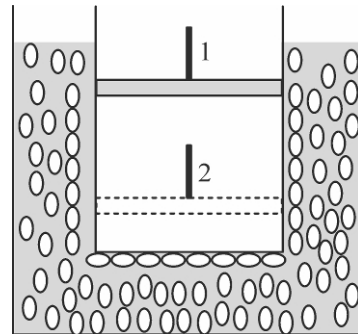
Get $V_2 = 50 \text{ m/sec}$ toward Indian submarine.

35.(B)
$$\lambda' = \frac{V \text{ wrt to observer}}{f'} = \frac{V + V_2}{f_0 \frac{(V + V_2)}{(V - 50)}} = \frac{V - 50}{f_0}$$

$$\lambda'' = \frac{V + 50}{f_0 \frac{(V + V_2)(V + 50)}{(V - V_2)(V - 50)}} = \frac{(V - V_2)(V - 50)}{f_0(V + V_2)}$$

$$\frac{\lambda'}{\lambda''} = \frac{V_1 + V_2}{V - V_2} = \frac{1050 + 50}{1050 - 50} = 1.1$$

- 36.(D) Since the piston is pushed very quickly from position 1 to position 2, the process AB is an adiabatic compression. Process BC is isochoric and process CA is an isothermal expansion. These are shown on the P-V diagram the P-V diagram correctly in option (D).



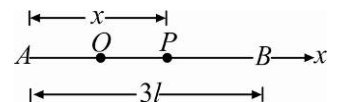
- 37.(A) If ΔW is the work done in the whole cycle, we have $\Delta Q = \Delta W + \Delta U$

But $\Delta U = 0$ for the entire cycle. Hence, $\Delta W = \Delta Q = mL = 100 \times 80 = 8000 \text{ cal} = 8 \text{ kcal}$

- 38.(C) Take x-axis along AB and origin at A. Let $AP = x$, then $PB = (3l - x)$

The force on the particle of mass m at P is $\vec{F}$ given by

$$\vec{F} = 2 \left(\frac{mg}{l} \right) (-\vec{x}) + \frac{mg}{l} \left(\frac{3l - x}{x} \right) \vec{x} = \left(3mg - \frac{3mg}{l} x \right) = \frac{3mg}{l} (l - x)$$



Hence, the equation of motion of mass m is: $m \frac{d^2 x}{dt^2} = \frac{3mg}{l} (l - x) = m\omega^2 (l - x)$

Where $\omega^2 = 3g / l$

This is a simple harmonic motion of period $T = 2\pi \sqrt{\frac{l}{3g}}$.

The mean position is at the point $x = l$ (i.e.) at O , where $AO = l$.

Now, for point A , displacement $= -l$ and hence $v_A^2 = \omega^2 [a^2 - (-l)^2] = \omega^2 (a^2 - l^2)$

Now, $v_A = 3\sqrt{gl}$ (given) and hence $v_A^2 = 9gl = \frac{3g}{l}(a^2 - l^2)$

$$a^2 = 4l^2 \quad \text{or} \quad a = 2l$$

39.(B) $OB = 2l$. O is the mean-position, B is the amplitude point. Also $|AP| = l = \frac{a}{2}$

Hence, the time t from A to O is given by $\frac{a}{2} = a \sin \omega t$

$$\text{Given } \omega t = \frac{\pi}{6} \Rightarrow t = \frac{\pi}{6\omega} = \frac{\pi}{6} \times \frac{T}{2\pi} = \frac{T}{12}$$

$$\text{Hence, time from } A \text{ to } B = \frac{T}{12} + \frac{T}{4} = \frac{T}{3}$$

40.(A) Since, B is the amplitude point, the instantaneous velocity at $B = v_B = 0$.

41.(B) For $r < R_T$, the conservation of energy implies that $4\pi r^2(-k) \frac{dT}{dr} = P \frac{4}{3} \pi r^3$

Therefore, the temperature difference to $37^\circ\text{C} = 310.15\text{K}$, $\Delta T(r)$, is given by $\Delta T(r) = -\frac{Pr^2}{6k} + C$

Where C is a constant

For $r > R_T$, the conservation of energy implies that $4\pi r^2(-k) \frac{dT}{dr} = P \frac{4}{3} \pi R_T^3$

Therefore, the temperature difference to 37°C is $\Delta T(r) = \frac{PR_T^3}{3kr}$

In this case there is no constant, since very far away the increase in temperature is zero.

Matching the two solutions at $r = R_T$ gives $C = \frac{PR_T^2}{2k}$

Therefore the temperature at the centre of the tumour, in SI units, is

$$\text{Temperature : } 310.15 + \frac{PR_T^2}{2k}$$

42.(A) The increase in temperature at the tumour surface (the lower temperature in the tumour) is

$$\Delta T(R_T) = \frac{PR_T^2}{3k}$$

This increase should be equal to 6.0 K . Therefore $P = \frac{3\Delta Tk}{R_T^2} = \frac{3 \times 6 \times 0.6}{0.05^2} = 4.3 \text{ kW} / \text{m}^3$

43.(A) Let $p(z)$ denote the pressure of air at height z ; then, according to one of the assumptions $p(z) = p(0) - \rho_{\text{Air}}gz$, where $p(0)$ is the atmospheric pressure at zero altitude.

Throughout the chimney the Bernoulli law applies, that is, we can write

$$\frac{1}{2} \rho_{\text{Smoke}} v(z)^2 + \rho_{\text{Smoke}}gz + p_{\text{Smoke}}(z) = \text{const.}, \quad (1)$$

where $p_{\text{Smoke}}(z)$ is the pressure of smoke at height z , ρ_{Smoke} is its density, and $v(z)$ denotes the velocity of smoke; here we have used the assumption that the density of smoke does not vary throughout the chimney. Now we apply this equation at two points, (i) in the furnace, that is at point $z = -\epsilon$, where ϵ is a negligibly small positive number, and (ii) at the top of the chimney where $z = h$ to obtain

$$\frac{1}{2} \rho_{\text{Smoke}} v(h)^2 + \rho_{\text{Smoke}}gh + p_{\text{Smoke}}(h) \approx p_{\text{Smoke}}(-\epsilon) \quad (2)$$

On the right hand side we have used the assumption that the velocity of gases in the furnace is negligible (and $-\rho_{Smoke}g\varepsilon \approx 0$).

We are interested in the minimal height at which the chimney will operate. The pressure of smoke at the top of the chimney has to be equal or larger than the pressure of air at altitude h ; for minimal height of the chimney we have $p_{Smoke}(h) \approx p(h)$. In the furnace we can use $p_{Smoke}(-\varepsilon) \approx p(0)$. The Bernoulli law applied in the furnace and at the top of the chimney (Eq. (2)) now reads.

$$\frac{1}{2}\rho_{Smoke}v(h)^2 + \rho_{Smoke}gh + p(h) \approx p(0) \quad (3)$$

$$\text{From this we get : } v(h) = \sqrt{2gh\left(\frac{\rho_{Air}}{\rho_{Smoke}} - 1\right)} \quad (4)$$

$$v(h) \geq \frac{B}{A} \quad (5)$$

$$\text{Form which we have } h \geq \frac{B^2}{A^2} \frac{1}{2g} \frac{1}{\frac{\rho_{Air}}{\rho_{Smoke}} - 1} \quad (6)$$

- 44.(C) We can treat the smoke in the furnace as an ideal gas (which is at atmospheric pressure $P(0)$ and temperature T_{smoke} . If the air was at the same temperature and pressure it would have the same density according to our assumptions. We can use this to relate the ratio $\rho_{Air} / \rho_{Smoke}$ to T_{Smoke} / T_{Air} , that is,

$$\frac{\rho_{Air}}{\rho_{Smoke}} = \frac{T_{Smoke}}{T_{Air}}, \text{ and finally} \quad (7)$$

$$h \geq \frac{B^2}{A^2} \frac{1}{2g} \frac{T_{Air}}{T_{Smoke} - T_{Air}} = \frac{B^2}{A^2} \frac{1}{2g} \frac{T_{Air}}{\Delta T} \quad (8)$$

For minimal height of the chimney we use the equality sign.

$$\frac{h(30)}{h(-30)} = \frac{(30+273)}{(400)-(30)} = \frac{(400)-(-30)}{(-30+273)}; h(30)=145m \quad (9)$$

- 45.(B) The velocity is constant,

$$v = \sqrt{2gh\left(\frac{\rho_{Air}}{\rho_{Smoke}} - 1\right)} = \sqrt{2gh\left(\frac{T_{Smoke}}{T_{Air}} - 1\right)} = \sqrt{2gh\frac{\Delta T}{T_{Air}}} \quad (10)$$

This can be seen from the equation of continuity $Av = \text{const.}$ (ρ_{Smoke} is constant). It has a sudden jump from approximately zero velocity to this constant value when the gases enter the chimney from the furnace. In fact, since the chimney operates at minimal height this constant is equal to B , that is $v = B/A$.

$$p_{Smoke}(z) = p(0) - (\rho_{Air} - \rho_{Smoke})gh - \rho_{Smoke}gz \quad (11)$$

Thus the pressure of smoke suddenly changes as it enters the chimney from the furnace and acquires velocity.

MATCH THE COLUMN TYPE

46. [A-R] [B-P] [C-P] [D-R]

$$PV^\gamma = \text{constant}, \quad V^{(\gamma-1)}T = \text{constant},$$

$$P^{(1-\gamma)}T^\gamma = \text{constant}$$

$$P \propto \frac{1}{V^\gamma}, P \propto T^{\gamma/(\gamma-1)}, V \propto T^{1/(\gamma-1)}$$

$$P \propto T^{\gamma/(\gamma-1)} \propto (I.E.)^{\gamma/(\gamma-1)}$$

47. [A-Q] [B-P] [C-R] [D-S]
- (A) $\lambda = L$ as $x = 0$ and $x = L$ are nodes
 $\therefore$ Terms should be either $\sin \frac{\pi x}{L}$ or $\sin \frac{2\pi x}{L}$ as $k = \frac{2\pi}{L} = \frac{2\pi}{\lambda} \Rightarrow \lambda = L$
 $\therefore$ (Q) is suitable for (A).
- (B) $\lambda = 2L$ as $x = 0$ and $x = L$ are nodes $\therefore$ Terms should be either $\sin \frac{\pi x}{L}$ or $\sin \frac{2\pi x}{L}$
 As $k = \frac{\pi}{L} = \frac{2\pi}{\lambda} \Rightarrow \lambda = 2L$ $\therefore$ (P) is suitable for (B)
- (C) $\lambda = 4L$ as $x = 0$ is node $\therefore$ Terms should be either $\cos \frac{\pi x}{2L}$ or $\cos \frac{2\pi x}{L}$
 As $k = \frac{2\pi}{\lambda} = \frac{\pi}{2L} \Rightarrow 4L$
 $\therefore$ (R) is suitable for (C)
 (D) $\lambda = 2L$ as $x = 0$ is node
 $\therefore$ Terms should be either $\cos \frac{\pi x}{L}$ or $\cos \frac{2\pi x}{2L}$
 As $k = \frac{\pi}{L} = \frac{2\pi}{\lambda} = \lambda = 2L$ $\therefore$ (S) is suitable for (D)
48. [A-Q] [B-R] [C-P] [D-S]
- (P) Pressure above surface $\times$ Area
 $\pi R^2 \times \rho gh$
- (B) Pressure below bottom $\times$ Area $= (2\rho gh + 2\rho gh)\pi R^2 = 4\pi R^2 \rho gh$
- (C) $F_V = \int_0^h \rho g(h+x) dx \times 2R = \rho Rg 3h^2$
 $F_r = \int_0^h (2\rho gh) + (2\rho gx) 2R dx$
 $= 4\rho gh \frac{3h^2}{2} \Rightarrow 6\rho Rgh^2$
 Total $= 9\rho Rgh^2$
- (D) $F_{Buoyancy} = \rho \alpha p R^2 hg + 2\rho \pi R^2 hg = 3\rho \pi R^2 gh$
49. [A-Q] [B-P] [C-S] [D-R] Analyse using circular motion analogy.
50. [A-Q] [B-S] [C-R] [D-R] Use : $\lambda' = \frac{V_{relative}}{f'}$

NUMERICAL/SUBJECTIVE TYPE

- 51.(2) As temperature changes by ΔT , new length l' and moment of inertia I' are given as :

$$l' = l(1 + \alpha \Delta T) \quad I' = I(1 + 2\alpha \Delta T)$$

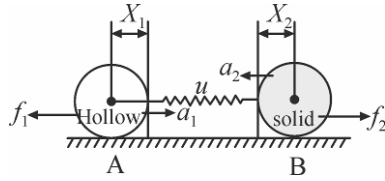
Thus new time period is

$$T' = 2\pi \sqrt{\frac{I'}{mgl'}} = 2\pi \sqrt{\frac{I}{mgl}} \frac{(1 + 2\alpha \Delta T)^{1/2}}{(1 + \alpha \Delta T)^{1/2}}$$

As $\alpha \Delta T \ll 1$, using binomial approximate we get

$$T' = 2\pi \sqrt{\frac{I}{mgl}} \left(1 + \frac{1}{2} \alpha \Delta T \right) \text{ change in period is } T' - T = 2\pi \sqrt{\frac{I}{mgl}} \left(\frac{1}{2} \alpha \Delta T \right) = \pi \alpha \Delta T \sqrt{\frac{I}{mgl}}$$

52.(21) x_1 & x_2 be the displacement from equilibrium position



Now for hollow sphere, applying $\tau_A = I_A \alpha$

$$k(x_1 + x_2)r = \frac{5}{3}mr^2 \frac{a_1}{r} \quad (i)$$

By angular momentum conservation (about A) of the system $\frac{5}{3}mv_1 r = \frac{7}{5}mv_2 r$

$$\Rightarrow 25v_1 = 21v_2 \quad (ii)$$

$$\Rightarrow 25x_1 = 21x_2 \quad (iii)$$

Using (i) and (iii) we get $k\left(x_1 + \frac{25}{21}x_1\right) = \frac{5}{3}m a_1 \Rightarrow a_1 = \frac{46}{35} \frac{k}{m} x_1$

Now for amplitude, $A_1 + A_2 = x$ (iv)

From equation (iii) we get $A_2 = \frac{25}{21}A_1$ (v)

By (iv) and (v) we get $A_1 = \frac{21}{46}x_0$

$$\text{Ans. } A_1 = \frac{21}{46}x_0 \quad ; \quad A_2 = \frac{25}{46}x_0 \quad ; \quad f = \frac{1}{2\pi} \sqrt{\frac{46k}{35m}}$$

53.(66.36) First resonance occurs when $v_{fork} = \frac{c}{4\ell_1}$; where ℓ_1 is the length of air column in the tube $\Rightarrow \ell_1 = \frac{1}{4}m$

Second resonance occurs when $v_{fork} = 3\frac{c}{4\ell_2} \Rightarrow \ell_2 = \frac{3}{4}m$

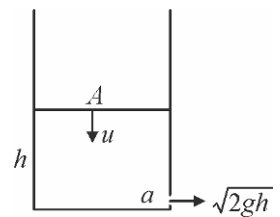
So, first resonance occurs when water level in the pipe is $1 - \frac{1}{4} = \frac{3}{4}m$ and second resonance at $1 - \frac{3}{4} = \frac{1}{4}m$.

When water level in h ,

$$Au = a\sqrt{2gh} \Rightarrow -\frac{dh}{dt} = \frac{a}{A}\sqrt{2gh}$$

$$\Rightarrow -\int_{\frac{3}{4}}^{\frac{1}{4}} \frac{dh}{\sqrt{h}} = \frac{a}{A}\sqrt{2g} \int_0^t dt \Rightarrow t = \frac{A}{a\sqrt{2g}} 2\left(\sqrt{\frac{3}{4}} - \sqrt{\frac{1}{4}}\right)$$

$$\Rightarrow t = \frac{\pi(2 \times 10^{-2})^2}{\pi(1 \times 10^{-3})^2 \sqrt{20}} (\sqrt{3} - 1) = \frac{400}{2\sqrt{5}} (1.73 - 1) = \frac{200(0.73)}{2.2} = \frac{730}{11} = 66.36s$$



54.(3) Given that from graph

$$\frac{1}{2}mV_m^2 = 15 \times 10^{-3} \Rightarrow V_m = \sqrt{0.150} \text{ m/s}$$

$$A\omega = \sqrt{0.150} \text{ m/s} \Rightarrow L\theta_m \sqrt{\frac{g}{L}} = \sqrt{0.150} \text{ m/s}$$

$$\Rightarrow \sqrt{gL} = \frac{\sqrt{0.150}}{100 \times 10^{-3}} \Rightarrow L = \frac{0.150}{0.1} = 1.5m$$

$$55. \quad \sin\left(30^\circ + \frac{\Delta\theta}{2}\right) = \frac{\frac{\ell}{2}(1 + \alpha_2 \Delta T)}{\ell(1 + \alpha_1 \Delta T)}$$

$$\sin 30^\circ \cos \frac{\Delta\theta}{2} + \cos 30^\circ \sin \frac{\Delta\theta}{2} = \frac{1}{2}(1 + \alpha_2 \Delta T)(1 + \alpha_1 \Delta T)^{-1}$$

$$\therefore \Delta\theta \text{ is very small : } \cos \frac{\Delta\theta}{2} \approx 1, \sin \frac{\Delta\theta}{2} = \frac{\Delta\theta}{2}$$

$$\Rightarrow \frac{1}{2}(1) + \frac{\sqrt{3}}{2} \left(\frac{\Delta\theta}{2}\right) = \frac{1}{2}[1 + (\alpha_2 - \alpha_1) \Delta T] \Rightarrow \Delta T = \frac{\sqrt{3} \Delta\theta}{2(\alpha_2 - \alpha_1)}$$

56. For equilibrium of piston, if T is tension in string and P_g is gas pressure

$$\text{For } m_1: P_0 s_1 + m_1 g + T = P_g S_1 \quad \dots (i); \quad \text{For } m_2: P_g S_2 + m_2 g = T + P_0 S_2 \quad \dots (ii)$$

$$(i) + (ii): P_0(s_1 - s_2) + (m_1 + m_2)g = P_g(s_1 - s_2)$$

$$P_g = P_0 + \frac{(m_1 + m_2)g}{\Delta s} \Rightarrow P_g \text{ remains same}$$

$$\text{By gas law, at temperature } T_1: P_g V = RT_1; \quad \text{At temp } T_2 (T_2 > T_1): P_g (V + \ell \Delta s) = RT_2$$

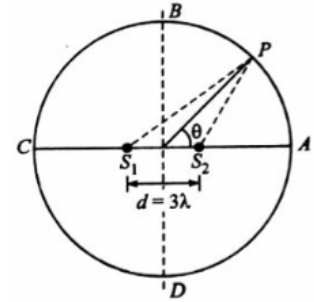
$$P_g \Delta s \ell = R(T_2 - T_1) \Rightarrow T_2 - T_1 = \frac{P_g \Delta s \ell}{R} = \frac{1}{R} [P_0 \Delta s + (m_1 + m_2)g]$$

57. For a point P on the circle shown in figure, the path difference in the two waves at P is $\Delta x = S_1 P - S_2 P = d \cos \theta = 3\lambda \cos \theta$

We know for constructive interference at P , the path difference must be an integral multiple of wavelength λ . Thus for a maxima at P , we have $3\lambda \cos \theta = 0$; $3\lambda \cos \theta = \lambda$; $3\lambda \cos \theta = 2\lambda$; $3\lambda \cos \theta = 3\lambda$

$$\text{or } \theta = \frac{\pi}{2} \text{ or } \theta = \cos^{-1} \frac{1}{3} \text{ or } \theta = \cos^{-1} \frac{2}{3} \text{ or } \theta = 0$$

There are total 12 points of maxima on the circle



58. Before expansion, $P_{in} = P_0 + \frac{4T}{r}$

$$\text{If charge } q \text{ is given to the soap bubble, its potential } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R}$$

After expansion of bubble at radius R , the inside air pressure will be

$$P_{in} = P_0 - \frac{\sigma^2}{2\epsilon_0} + \frac{4T}{R} \quad \left[\text{here } \sigma = \frac{q}{4\pi R^2} = \frac{\epsilon_0 V R}{R^2} = \frac{\epsilon_0 V}{R} \right]$$

$$\text{By gas law } P_1 V_1 = P_2 V_2$$

$$\left(P_0 + \frac{4T}{r} \right) \left(\frac{4}{3} \pi r^3 \right) = \left(P_0 - \frac{\sigma^2}{2\epsilon_0} + \frac{4T}{R} \right) \left(\frac{4}{3} \pi R^3 \right)$$

$$\Rightarrow P_0(R^3 - r^3) + 4T(R^2 - r^2) - \frac{\epsilon_0 V^2 R}{2} = 0$$

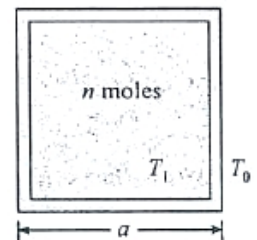
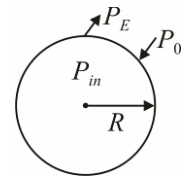
59. As shown in figure if at time $t = 0$, gas temperature is T_1 and after time t , its temperature is T , heat will be conducted through the walls of container, the rate of heat conducted to outside surrounding can be written as

$$\frac{dQ}{dt} = \frac{k(6a^2)}{x} [T - T_0] \quad \dots (i)$$

If dQ heat is conducted outside in a time dt and due to this if the temperature of gas falls by dT , we have

$$dQ = -nC_v dt \quad [\text{As volume of container is constant}]$$

$$\text{or } dQ = -\frac{nR}{\gamma - 1} dT \quad [\text{as } C_v = \frac{R}{\gamma - 1}] \quad \dots (ii)$$



From equation (i) and (ii) we have

$$\frac{nR}{\gamma-1} \frac{dT}{dt} = -\frac{6ka^2}{x}(T-T_0) \text{ or } \frac{dT}{T-T_0} = -\frac{6ka^2(\gamma-1)}{nRx} dt$$

Integration the above expression within proper limits from beginning ($t = 0$), we get

$$T = T_0 + (T_1 - T_0)e^{-\frac{6ka^2(\gamma-1)}{nRx}t}$$

60. (i) According to the first law of thermodynamics, the amount of heat transmitted dQ to the gas in the bubble is found as $dQ = nC_V dT + p dV$

$$\text{where the molar heat capacity at arbitrary process is as follows } C = \frac{1}{n} \frac{dQ}{dT} = C_V + \frac{p}{n} \frac{dV}{dT}$$

Here C_V stands for the molar heat capacity of the gas at constant volume, p designates its pressure, n is the total amount of moles of gas in the bubble, V and T denote the volume and temperature of the gas, respectively.

According to the Laplace formula, the gas pressure inside the bubble is defined by $p = \frac{4\sigma}{r}$

Thus, the equation of any equilibrium process with the gas in the bubble is a polytrope of the form

$$p^3 V = \text{const}$$

The equation of state of an ideal gas has the form $pV = nRT$ and hence equation $T^3 V^{-2} = \text{const}$

Differentiating the derivative with respect to temperature sought is found as $\frac{dV}{dT} = \frac{3V}{2T}$

Taking into account that the molar heat capacity of a diatomic gas at constant volume is $C_V = \frac{5}{2}R$

and using it is finally obtained that $C = C_V + \frac{3}{2}R = 4R$

- (ii) Since the heat capacity of the gas is much smaller than the heat capacity of the soap film, and there is heat exchange between them, the gas can be considered as isothermal since the soap film plays the role of thermostat. Consider the fragment of soap film, limited by the angle α as shown in the figure.

It's area is found as : $S = \pi(ar)^2$

And the corresponding mass is obtained as $m = \rho Sh$.

Let x be an increase in the radius of the bubble, then the Newton second law for the fragment of the soap film mentioned above takes the form $m\ddot{x} = p'S' - F_{surf}$.

Where F_{surf} denotes the projection of the resultant surface tension force acting in the radial direction,

p' stands for the gas pressure beneath the surface of the soap film and $S' = S\left(1 + 2\frac{x}{r}\right)$.

F_{surf} is easily found as $F_{surf} = F_{ST} \alpha = \sigma \cdot 2 \cdot 2\pi[r+x]\alpha \cdot \alpha$

Since the gaseous process can be considered isothermal, it is written that

$$p'V' = pV$$

Assuming that the volume increase is quite small,

$$p' = p \frac{1}{\left(1 + \frac{x}{r}\right)^3} \approx p \frac{1}{\left(1 + \frac{3x}{r}\right)} \approx p \left(1 - \frac{3x}{r}\right)$$

Thus, from (B . 10) – (B . 16) and (B . 3) the equation of small oscillations of the soap film is derived

$$\text{as } \rho h \ddot{x} = -\frac{8\sigma}{r^2} x. \text{ With the frequency } \omega = \sqrt{\frac{8\sigma}{\rho h r^2}}$$

